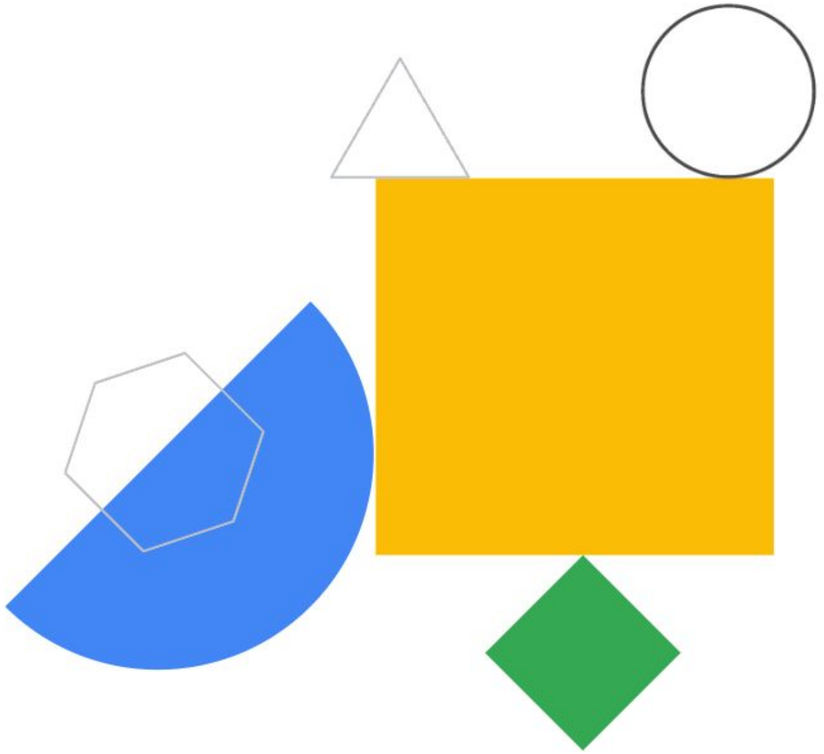
Google Cloud

Investigating Application Performance Issues



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Objectives

01 Explain the features and benefits of Error Reporting, Cloud Trace, and Cloud Profiler.

02 List and explain the functionalities of Error Reporting, Cloud Trace, and Cloud Profiler.



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In this section, you explore



- Error Reporting
- Cloud Trace
- Cloud Profiler

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Error Reporting overview



Counts, analyzes, and aggregates the **crashes** to report them on your preferred notification channel.



Error Reporting can only analyze log entries that:

- Are **stored in Cloud Logging buckets** in the **global region.**
- Have the same **source and destination** Google Cloud **projects.**
- Have the **customer-managed encryption keys** (CMEK) **disabled.**

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Error Reporting features

Understand errors

Error reporting brings processed data directly that help fix the root causes faster.

Automatic and real-time

An interface with sorting and filtering capabilities to report errors.

Instant error notification

Instantly alerts you when a new application error cannot be grouped with existing one.

Broad language and product support

- App Engine standard and flexible environments
- Cloud Run functions
- Apps Script
- Cloud Run
- Compute Engine
- Amazon EC2
- GKE

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Setting up Error Reporting

Setup process depends on the language and compute environments.

- Assign the **Error Reporting Writer IAM** role.
- Enable the Error Reporting API.
- Install the client library.

```
npm install --save @google-cloud/error-reporting
```

01 Perform the initial setup

02 Instrument Node.js apps

03 Configure environments to report error data

04 View error reports

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Setting up Error Reporting for a Node.js application

```
//Import the library
const {ErrorReporting} = require('@google-cloud/error-reporting');

// Instantiates a client
const errors = new ErrorReporting();

// Report errors
const errorEvent = errors.event();

// Add error information
errorEvent.setMessage('My error message');
errorEvent.setUser('root@nexus');

// Report the error event
errors.report(errorEvent, () => {
  console.log('Done reporting error event!');
});
```

01 Perform the initial setup

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Setting up Error Reporting for a Node.js application

 Cloud Run functions, Cloud Run, App Script, App Engine	Error reporting is automatically enabled.
 GKE	Add `cloud-platform` access scope during cluster creation.
 Compute Engine	Ensure that the service account used has the Error Reporting Writer role.
Outside Google Cloud	Provide the Google Cloud project ID and service account credentials to the Error Reporting library.

01 Perform the initial setup

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View error reports

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By default, recently occurring open and acknowledged errors is shown in order of **frequency**.

Add an issue tracker link to an error.

Error Repor

Error Group

Configu

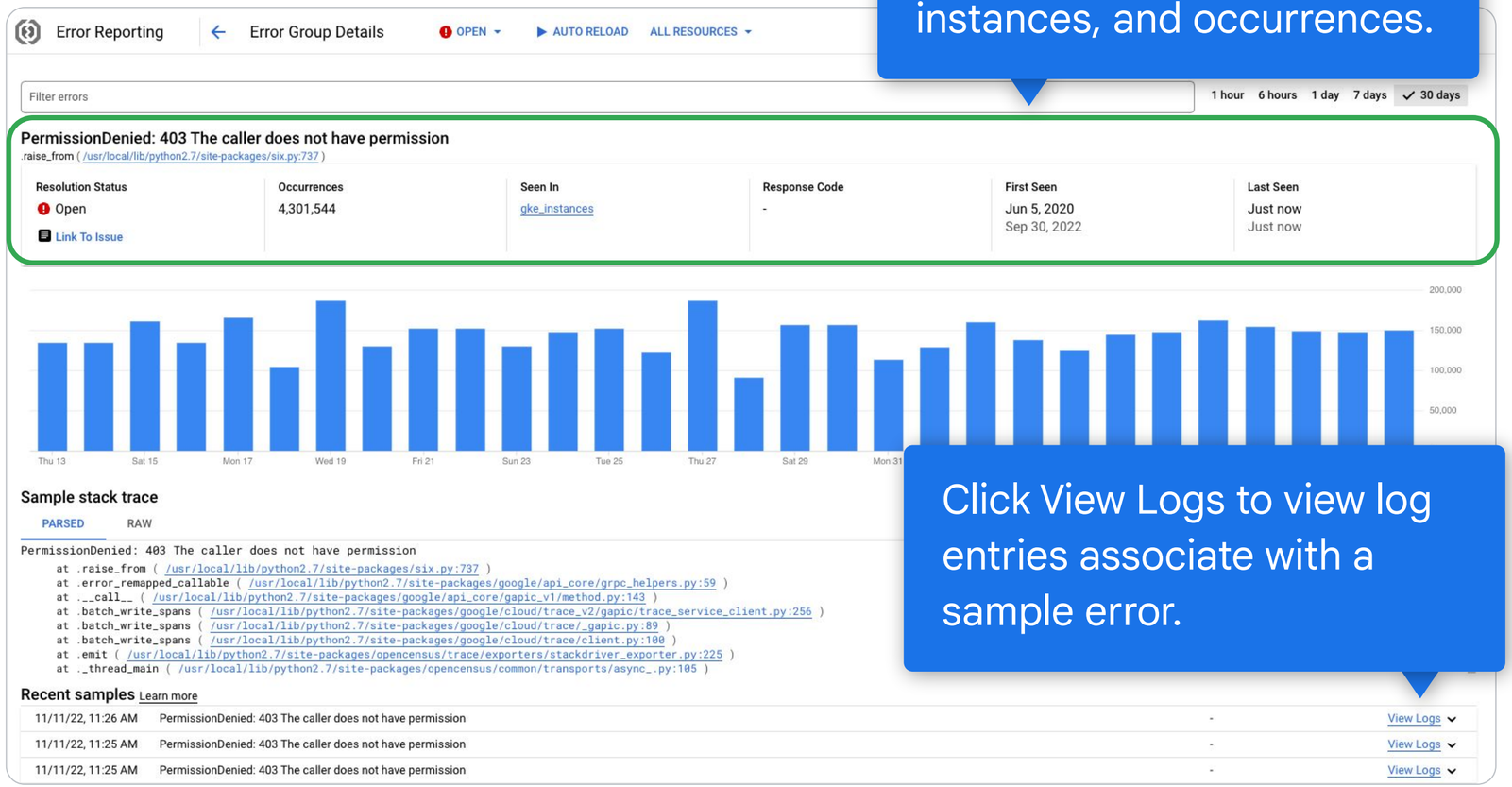
Filter errors

6 hours

Resolution Status	Occurrences ↓	Error	Seen In	Type	First Seen	Last Seen	Response Code	Issue Link
OPEN	57,187 57,187	Back-off restarting failed container	k8s_pods	Service error Troubleshooting guide	Mar 12, 2022 Mar 27, 2023	Just now Just now	-	+
OPEN	20,023 20,023	Liveness probe failed: Get "https://10.12.0.11:10250/livez": net/http: request canceled ...	k8s_pods	Service error Troubleshooting guide	Nov 15, 2021 Mar 27, 2023	4 minutes ago 4 minutes ago	-	+
OPEN	17,786 17,786	pod didn't trigger scale-up:	k8s_pods	Service error Troubleshooting guide	Dec 7, 2021 Mar 12, 2023	4 minutes ago 4 minutes ago	-	+
OPEN	9,690 9,690	MountVolume.SetUp failed for volume "volume-serving-cert" : secret "cm-adapter-servin...	k8s_pods	Service error Troubleshooting guide	Aug 9, 2022 Mar 28, 2023	1 minute ago 1 minute ago	-	+
OPEN	175 175	network is not ready: container runtime network not ready: NetworkReady=false reason:...	k8s_pods	Service error Troubleshooting guide	Jul 29, 2022 Apr 1, 2023	5 days ago 5 days ago	-	+
OPEN	6 6	Error: ErrImagePull	k8s_pods	Service error Troubleshooting guide	Mar 1, 2023 Mar 22, 2023	Apr 12, 2023 Apr 12, 2023	-	+
OPEN	6 6	Error: ImagePullBackOff	k8s_pods	Service error Troubleshooting guide	Mar 1, 2023 Mar 1, 2023	Apr 12, 2023 Apr 12, 2023	-	+
OPEN	2 2	Http2Exception$StreamException: Stream closed before write could take place 173 Http2Exception.java(io.netty.handler.codec.http2.Http2Exception.streamError)	adservice:adserv ...7b6c69-fk6z2	Application error	Apr 13, 2023 Apr 13, 2023	Apr 13, 2023 Apr 13, 2023	-	+
OPEN	1 1	The request was aborted because there was no available instance. Additional troublesh...	adservice-5457ff7:adserv ...ff7-00001-zbr	Service error Troubleshooting guide	7 days ago 7 days ago	7 days ago 7 days ago	-	+

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View error details



01 Perform the initial setup

02 Instrument Node.js apps

03 Configure environments to report error data

04 View error reports

Google Cloud

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In this section, you explore



- Error Reporting
- **Cloud Trace**
- Cloud Profiler

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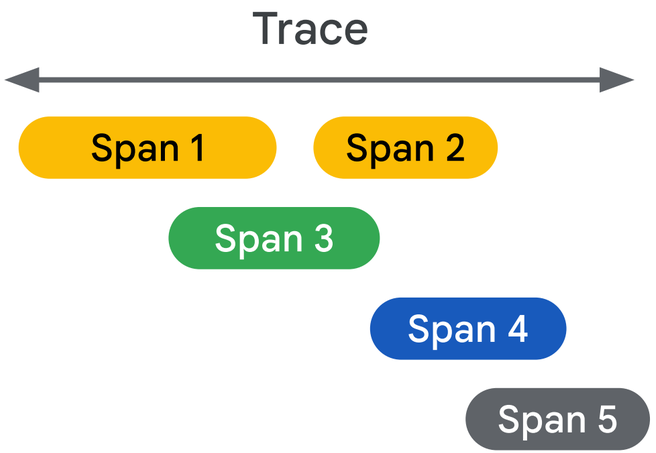
Cloud Trace tracks application latency

What is Cloud Trace?	It's a distributed tracing system that collects latency data from your applications and displays it in the Google Cloud console.
How does it do it?	• By continuously gathering and analyzing trace data to identify changes in performance • By reporting latency data on per URL basis and helping identify bottlenecks.
Platform support	• App Engine standard and flexible environments • Cloud Run • Cloud Run functions • Compute Engine • GKE • Non Google Cloud environment

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Cloud Trace terminologies

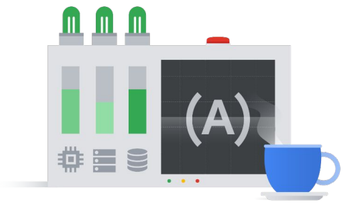
Tracing client	Collects spans and sends them to Cloud Trace.
Trace	Trace describes the time it takes an application to complete a single operation.
Span	How long it takes to perform a complete sub-operation.



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Sending trace data to Cloud Trace

Automatic tracing



- App Engine standard environment with Java 8, Python 2, and PHP 5 applications
- HTTP requests and latency data from Cloud Run functions and Cloud Run

Instrumenting the application



- Using Google client libraries or OpenTelemetry (recommended)

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Required IAM permissions



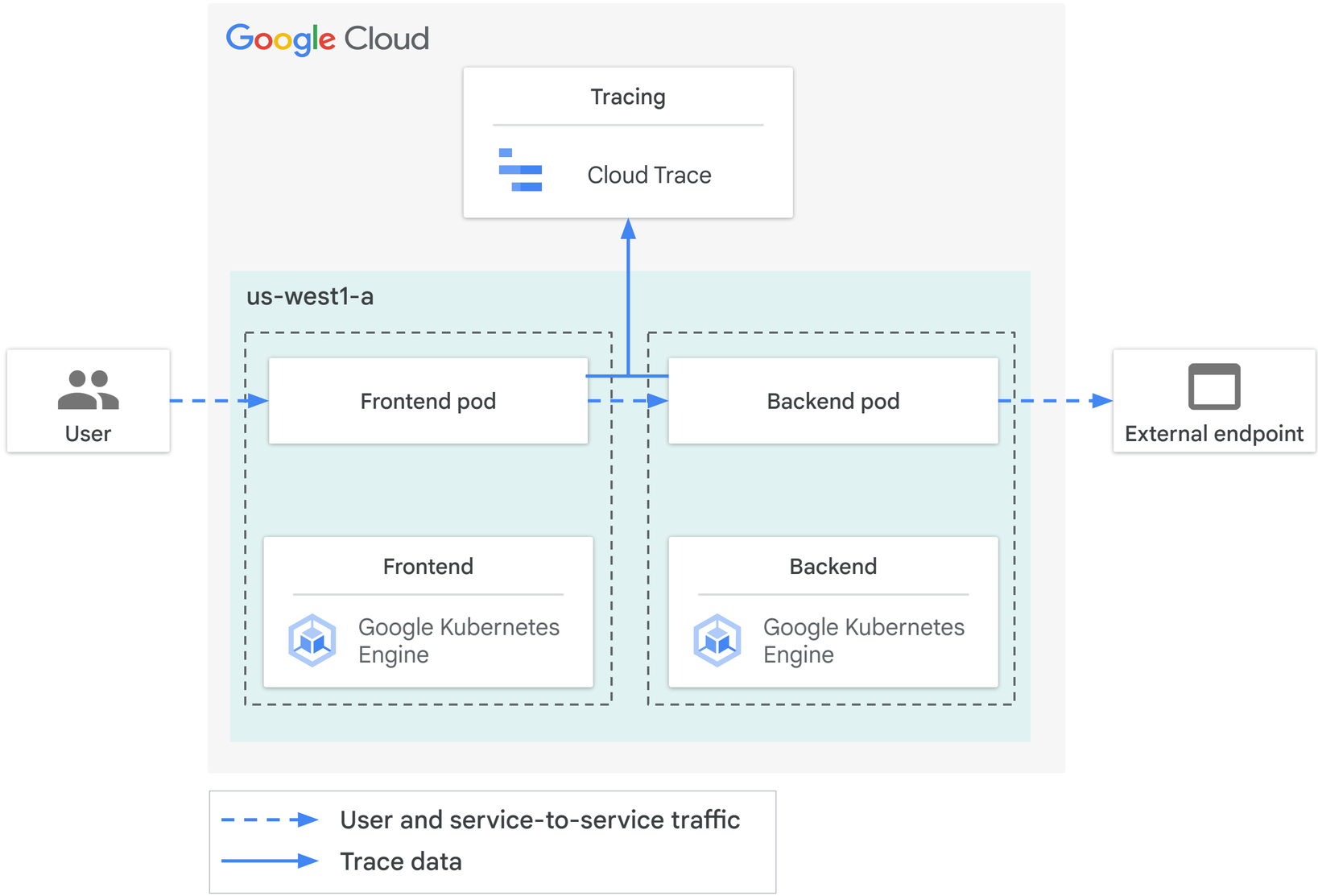
Cloud **Trace Agent role** is needed to send trace data from the application to Cloud Trace.

These products have default access:

- App Engine
- Cloud Run
- Cloud Run functions
- Google Kubernetes Engine
- Compute Engine*

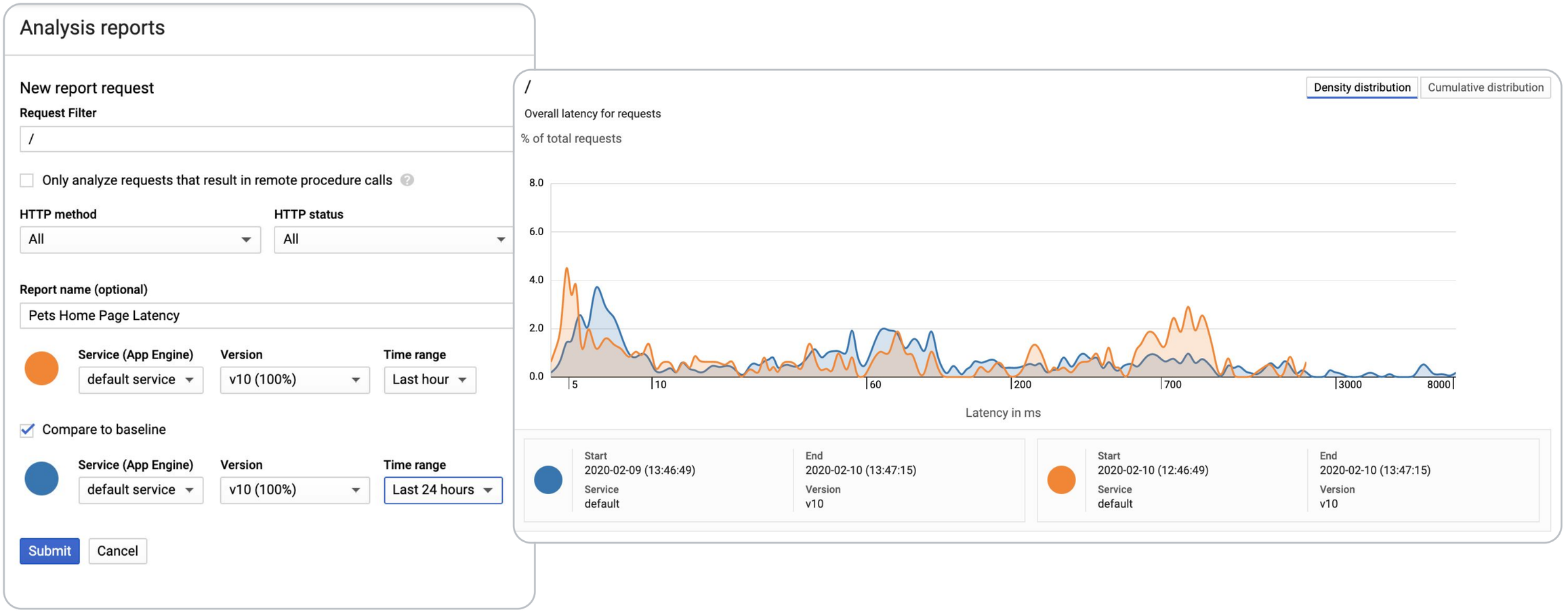
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Use case: Observing microservice latency with Cloud Trace



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Analysis reports show requests over time and compares versions and timespans

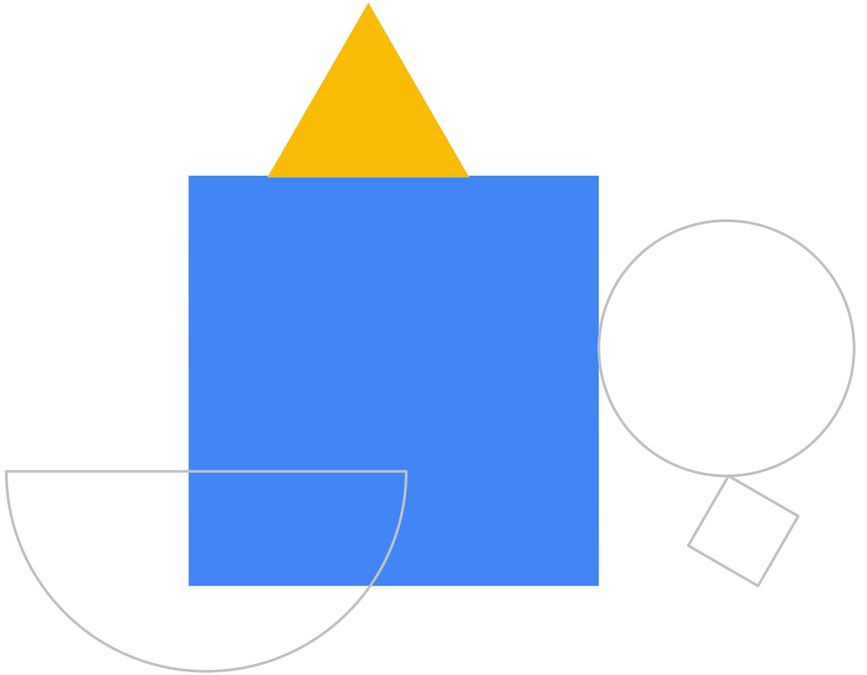


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Lab Intro

View application latency with Cloud Trace



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In this section, you explore



- Error Reporting
- Cloud Trace
- **Cloud Profiler**

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Cloud Profiler overview



Cloud Profiler

- Continuous profiling of production systems
- Statistical, low-overhead memory, and CPU profiler
- Insights contextualized to your code

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Cloud Profiler features

Low-impact production profiling

Provides a comprehensive application performance view without slowing it down.

Practical application profile creation

Continually analyzes the performance of CPU or memory-intensive functions running in an application.

Broad language and product support

- App Engine standard and flexible environments
- Compute Engine
- GKE
- Non-Google Cloud environment

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Available profiles

Profile type	Go	Java	Node.js	Python
CPU time	Y	Y		Y
Heap	Y	Y	Y	
Allocated heap	Y			
Contention	Y			
Threads	Y			
Wall time		Y	Y	Y

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Supported compute technologies

Environment	Go	Java	Node.js	Python
Compute Engine	Y	Y	Y	Y
Google Kubernetes Engine	Y	Y	Y	Y
App Engine flexible environment	Y	Y	Y	Y
App Engine standard environment	Y	Y	Y	Y
Dataproc		Y		
Dataflow		Y		Y
Outside of Google Cloud	Y	Y	Y	Y

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Setting up the Profiler agent in your code

```
gcloud services enable cloudprofiler.googleapis.com
```

Enable the Profiler API

```
sudo apt-get install -y build-essential
sudo apt-get install -y python3-pip
```

Install compiler and development tools

```
pip3 install google-cloud-profiler
```

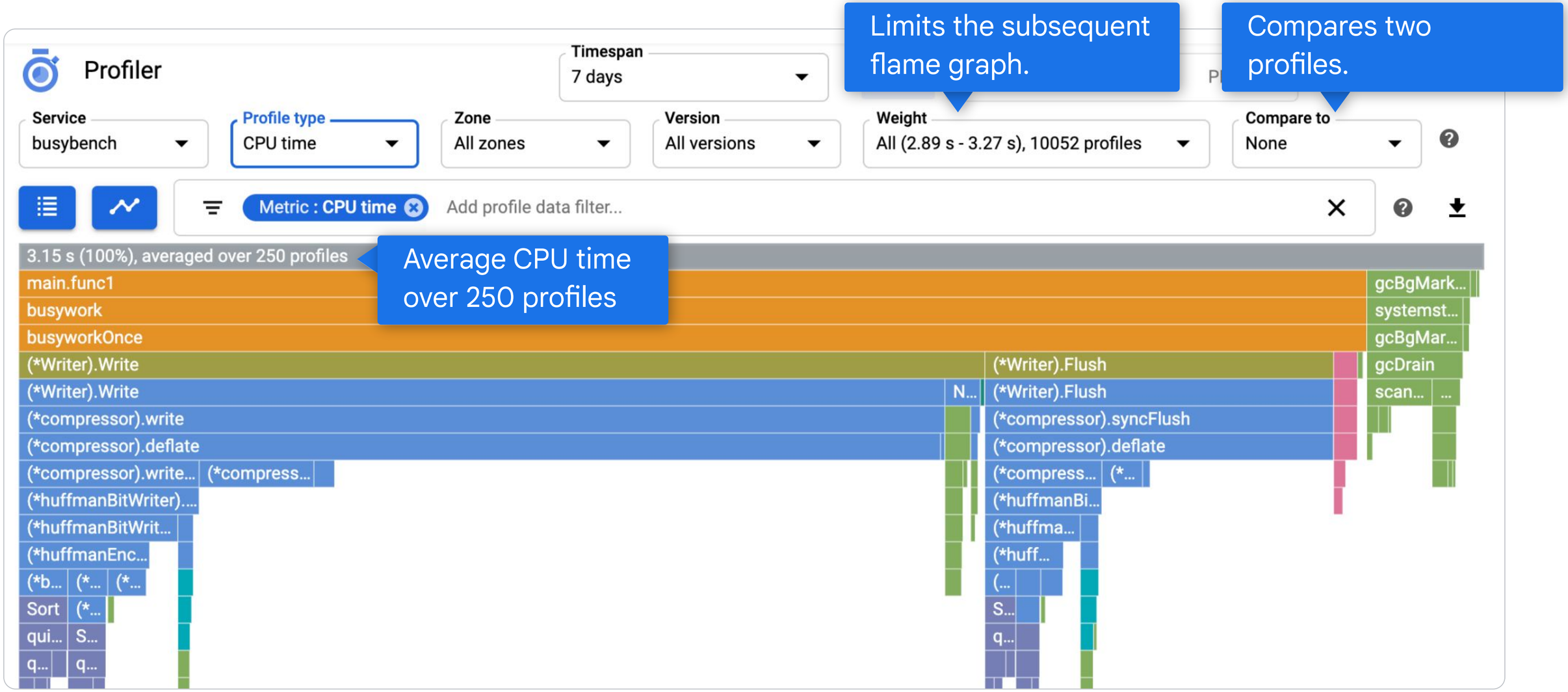
Install Profiler package

```
import googlecloudprofiler
def main():
  try:
    googlecloudprofiler.start(verbose=3)
  except (ValueError, NotImplementedError) as exc:
    print(exc)  # Handle errors here
```

Import **googlecloudprofiler** module and call the start function

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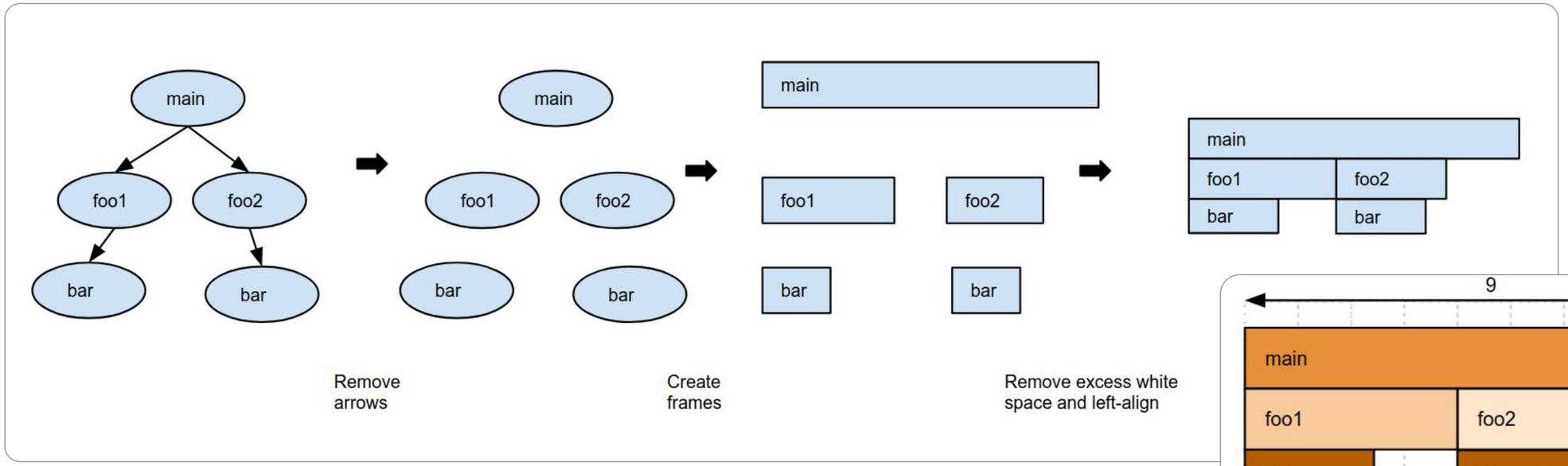
Profiler interface

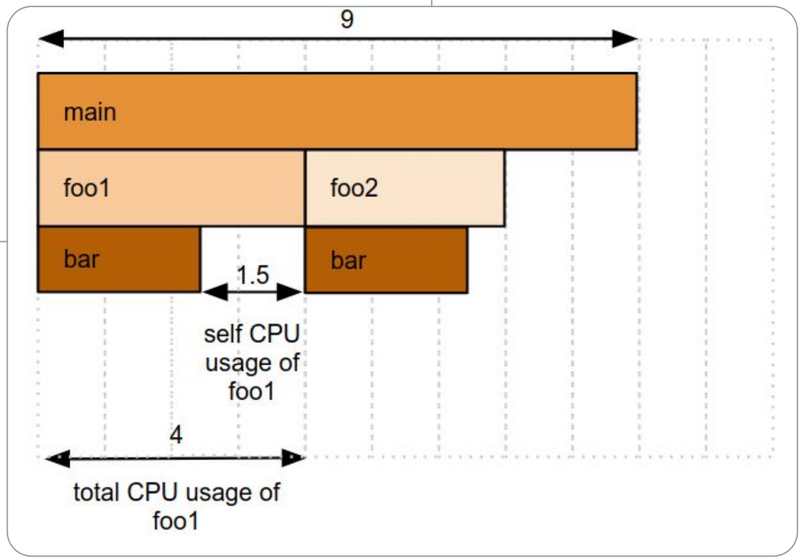


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Flame graph 101

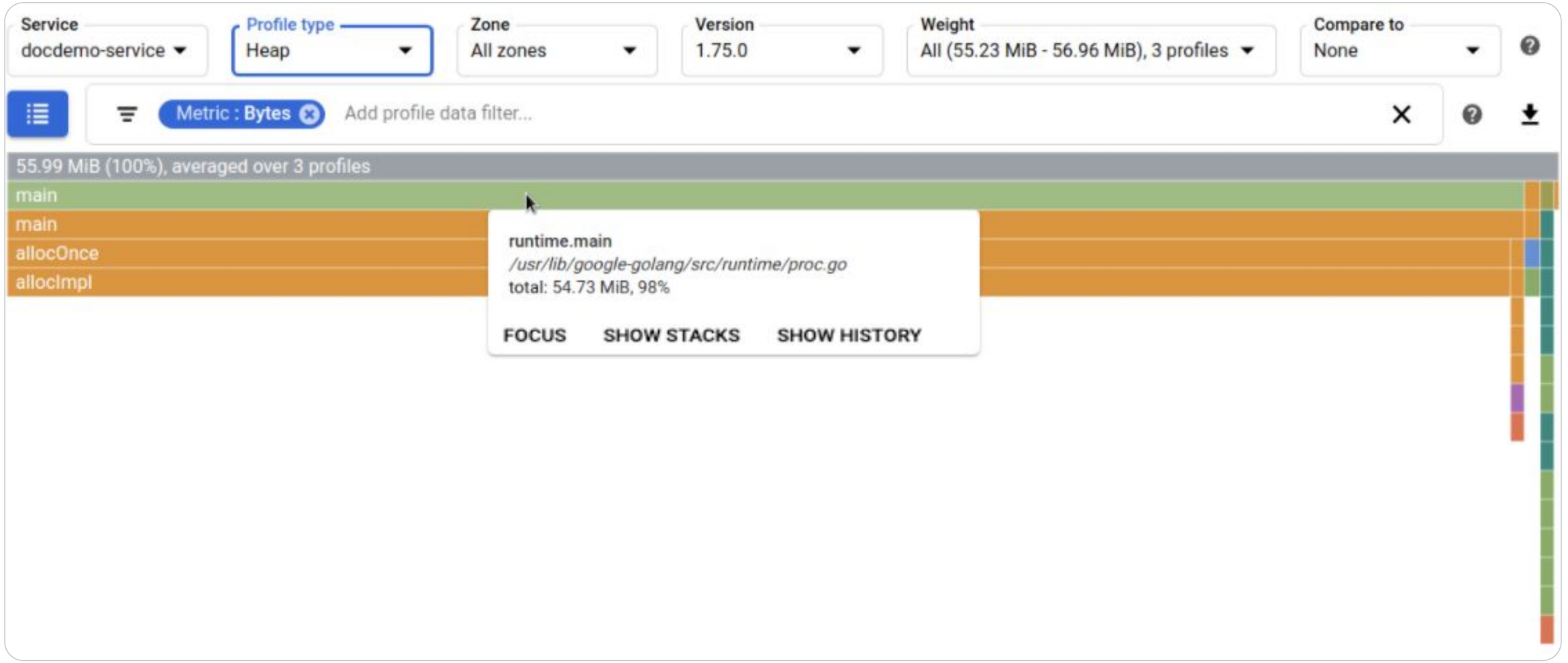




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Point to a frame



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Selecting a frame



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Knowledge Check



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Quiz | Question 1

Question

You have an SLO that states that 90% of your http requests need to respond in less than 100 ms. You want a report that compares latency for your last two versions. What tool would you use to most easily create this report?

A. Profiler

B. Trace

C. Error Reporting

D. Logging

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Quiz | Question 1

Answer

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B. Trace 

C. Error Reporting

D. Logging

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Quiz | Question 2

Question

You deployed a new version of a service and all of a sudden significantly more instances are being created in your Kubernetes cluster. Your service scales when average CPU utilization is greater than 70%. What tool can help you investigate the problem?

A. Profiler

B. Trace

C. Error Reporting

D. Logging

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Quiz | Question 2

Answer

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B. Trace

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Quiz | Question 3

Question

Which profile signifies the amount of memory allocated in the heap of the program?

A. Heap

B. Allocated heap

C. CPU time

D. Wall time

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Quiz | Question 3

Answer

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Recap

01 Explain the features and benefits of Error Reporting, Cloud Trace, and Cloud Profiler.

02 List and explain the functionalities of Error Reporting, Cloud Trace, and Cloud Profiler.



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